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StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2026 Jan-.

Rheumatoid Arthritis

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Last Update: May 25, 2023.

Continuing Education Activity

Rheumatoid arthritis (RA) is a systemic autoimmune disease characterized by inflammatory arthritis and extra-articular involvement. It is a chronic inflammatory disorder caused in many cases by the interaction between genes and environmental factors, including tobacco, that primarily involves synovial joints. It typically starts in small peripheral joints, is usually symmetric, and progresses to involve proximal joints if left untreated. Joint inflammation over time leads to the destruction of the joint with loss of cartilage and bone erosions. RA with a symptom duration of fewer than six months is defined as early RA, and when the symptoms have been present for more than six months, it is defined as established RA. RA, if untreated, is a progressive disease with morbidity and increased mortality. This activity describes the evaluation and management of rheumatoid arthritis and reviews the role of the interprofessional team in improving care for patients with this condition.

Objectives:

- Identify the comorbid conditions associated with rheumatoid arthritis.
- Summarize the pathophysiology of rheumatoid arthritis.
- Outline the use of rheumatoid factor and anti-citrullinated protein antibodies in evaluating rheumatoid arthritis.
- Review the importance of improving care coordination among the interprofessional team to improve outcomes for those affected with rheumatoid arthritis.

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Introduction

Rheumatoid arthritis (RA) is a systemic autoimmune disease characterized by inflammatory arthritis and extra-articular involvement. It is a chronic inflammatory disorder caused in many cases by the interaction between genes and environmental factors, including tobacco, that primarily involves synovial joints.[1] It typically starts in small peripheral joints, is usually symmetric, and progresses to involve proximal joints if left untreated.[2][3][4] Joint inflammation over time leads to the destruction of the joint with loss of cartilage and bone erosions. RA with a symptom duration of fewer than six months is defined as early RA, and when the symptoms have been present for more than six months, it is defined as established RA. [3] RA, if untreated, is a progressive disease with morbidity and increased mortality.[5]

There is no pathognomonic laboratory test for rheumatoid arthritis, which makes the diagnosis of this disease challenging in the early stages. A comprehensive clinical approach is required to make the diagnosis and prevent debilitating joint damage.[2][3] The treatment of patients with rheumatoid arthritis requires both pharmacological and non-pharmacological therapy. Today, the standard of care is early treatment with disease-modifying anti-rheumatic drugs. Despite treatment, many patients progress to disability and suffer significant morbidity over time. A comprehensive pharmacological and non-pharmacological treatment (physical therapy, counseling, and patient education) is required to improve clinical outcomes.[4]

The etiology of RA has a significant basis in genetics. It is thought to result from the interaction between patients' genotypes and environmental factors. In a nationwide study of 91 monozygotic (MZ) and 112 dizygotic (DZ) twin pairs in the United Kingdom, the overall MZ concordance rate was 15%, and in dizygotic twins, 5%.[6] The heritability of rheumatoid arthritis is approximately 40% to 65% for seropositive rheumatoid arthritis and 20% for seronegative rheumatoid arthritis.[6] The risk of developing rheumatoid arthritis has been associated with HLA-DRB1 alleles: *HLA-DRB1*04*, *HLA-DRB1*01*, and *HLA-DRB1*10*. These HLA-DRB1 alleles contain a stretch of a conserved sequence of 5 amino acids referred to as the "shared epitope" (SE) in the third hypervariable region of their DRB1 chain, which has been associated with the risk of developing RA.[7][8][9][10]

Polymorphisms in other genes are associated with RA, including *PADI4*, *PTPN22*, *CTLA4*, *IL-2RA*, *STAT4*, *TRAF1*, *CCR6*, and *IRF5*.^{[11][12][13]} Single nucleotide polymorphism (SNP) in *PSORS1C1*, *PTPN22*, and *MIR146A* genes are associated with severe disease.^[14] Some genetic polymorphisms are associated with RA in different ethnic groups.^{[12][14]}

The term epigenetics refers to heritable changes without altering the DNA sequence. These changes may be present in chromatin or the DNA. These include DNA methylation, histone modification, and non-coding RNA-mediated regulation. RA-FLS (fibroblast-like synoviocytes) overexpress tyrosine phosphatase SHP-2, coded by gene PTPN11, compared to synoviocytes from osteoarthritis (OA) patients, promoting the invasive nature of RA-FLS. The enhancer region of the PTPN11 intron contained two hypermethylated sites, resulting in abnormal epigenetic regulation of the gene and alteration of the function of RA-FLS.[15]

Cigarette smoking is the strongest environmental risk factor associated with rheumatoid arthritis. Studies have shown in anti-citrullinated protein antibody (anti-CCP) positive individuals, there is an interaction between the shared epitope (SE) and smoking that increases the risk of RA.[16][17][18][19][20][21][22]

Other environmental triggers may play a role as a trigger for RA, which is more closely associated with seropositive RA. These include silica, asbestos, textile dust, and *P gingivalis*. [1] This suggests that external exposure to various antigens in parts of the host distant from the joints then triggers an autoimmune inflammatory response in the joints. These distant locations include the lungs, oropharynx, and GI tract.[1] Changes in the composition and function of the intestinal microbiome have been related to rheumatoid arthritis as well. The composition of the gut microbiome becomes altered in patients with rheumatoid arthritis (dysbiosis), where rheumatoid arthritis patients have decreased gut microbiome diversity compared with healthy individuals. There is an increase in these genera: *Actinobacteria*, *Collinsella*, *Eggerthalla*, and *Faecalibacterium*. *Collinsella* alters gut mucosal permeability and has been related to increased rheumatoid arthritis disease severity.[22]

Genes Associated with Increased Risk of RA [11] [12]

Gene
product

Gene

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Table

Environmental Factors Associated with an Increased Risk of Developing RA [1]

Table



Epidemiology

The worldwide prevalence of RA in the Global Burden of Disease 2010 Study is about 0.24%. [23] The prevalence of RA is higher in Western and Northern Europe, North America, and other regions with people of European descent, such as Australia. The prevalence is lower in Central and South America and even lower in East Asia and Africa.[23] The annual incidence of RA in the United States and other western nations of northern Europe is about 40 per 100,000 persons. [24] According to epidemiologic data, RA is more prevalent in women compared to men, with a lifetime risk of RA of 3.6% in women compared to 1.7% in men.[25] RA risk also increases with age, with a peak incidence between age 65 to 80 years of age.[26][27][26] A systematic review of population-based studies (including 60 studies) showed a worldwide period prevalence of RA of 0.51% (1955-2015).[28]

The period prevalence of RA was more in urban areas (0.69%) than in rural areas (0.54%). The period prevalence of RA was higher in high-income countries (0.49%) compared to low-income countries (0.35.%).[29] RA prevalence was higher in North America and Europe and lower in Asia and South America (See Table below).[28] Data from Africa is limited and varies significantly by country, with RA being more common in North Africa and lower in sub-Saharan Africa.[29] RA is much less prevalent in East Asia compared to the rest of Asia.[29] There was an increase in the global period prevalence of 9.75% over time between 1980-2019 in a meta-analysis of 67 studies.[29]

As outlined above, there is a genetic disposition towards RA, which was demonstrated to be about 40% in a large study from Sweden in 2013.[30] The study also reported a higher heritability for seropositive RA and early-onset RA. According to their report, the risk of RA with a first-degree relative positive for RA is three-fold higher than a second-degree relative with RA giving a two-fold higher risk.[30] Multiple different genetic predispositions to explain this finding have now been identified. The strongest genetic predisposition for RA is from the HLA-DRB1 region (shared epitope).[7]

Among modifiable risk factors, cigarette smoking has the strongest association with RA. [21] Diet and nutrition have been shown to play a significant role as environmental triggers for RA. The typical 'western' diet that is rich, high in caloric content, and low in fiber increases the risk of RA.[31] Consumption of long-chain omega-3 polyunsaturated fatty acids is associated with a reduced risk of RA.[31]

Obesity is another well-established risk factor for RA. There is a 30% increase in the risk of RA for patients with a body mass index (BMI) of greater than 30 kg/m² and a 15% increased risk for those with a BMI of 25 to 29.9 kg/m². [32]

There is significant literature regarding the association of RA in patients with chronic mucosal or periodontal disease. However, no clear, consistent link has been identified in well-established studies. There is evidence that mucosal injury from occupational exposures and environmental pollutants can increase the risk of RA.[33][34][1][34]

Prevalence of RA (period prevalence 1955-2015)[29]

Continent/Number of Studies/RA Pooled Prevalence (95%CI)

Table

Pathophysiology

RA, in some patients, is triggered by some sort of environmental factor in a genetically predisposed host. The best example is tobacco use in a patient with the HLA-DRB1 "shared epitope" gene and the development of ACPA-positive RA.[1] RF and ACPA antibodies are the best-known autoantibodies in RA, but several other autoantibodies are relatively specific for RA. The presence of antibodies in rheumatoid arthritis is referred to as seropositive RA. RF is an antibody of any isotype that binds to the Fc portion of IgG.

RA patients often have antibodies to citrullinated proteins. These antibodies have been identified in patients with RA since 1964 (antiperinuclear factor)[35] and were also described in 1979 (anti-keratin antibodies)[36] by different assays. In the 1990s, these antibodies were determined to be the same antibodies with high specificity for RA.[37] The antibodies were found to have specificity for filaggrin, a citrullinated peptide.[38][37] The epitope for these antibodies is citrullinated peptides. A cyclic citrullinated peptide (CCP) was synthesized, which could be used in an ELISA to test for these antibodies in patients in a clinical situation.[39] These antibodies are called anti-cyclic citrullinated peptide antibodies (ACPA). Citrulline is derived from the post-transcriptional modification of arginine by peptidyl arginine deiminase (PAD). This reaction occurs at sites of tissue damage and inflammation, such as the lungs in smokers. HLA-DRB1 shared epitope preferentially presents epitopes containing citrulline. ACPA can be IgG, IgM, or IgA isotypes. ACPA can bind citrullinated residues on self-proteins like vimentin, fibronectin, fibrinogen, histones, and type 2 collagen.[22]

Anti-carbamylated protein antibodies(anti-CarP antibodies) are also found in patients with RA. Carbamylation is the conversion of lysine to homocitrulline in the presence of urea and cyanate. Myeloperoxidase converts thiocyanate to cyanate. The molecular structure of homocitrulline is similar to citrulline; however, anti-CarP antibodies are distinct antibodies that have been associated with RA in both ACPA-positive and ACPA-negative patients.[22] There are other autoantibodies that have been described in RA patients, including those directed against fibrinogen, enolase, and vimentin.[40]

The immune response in RA starts at sites distant from the synovial joints, such as the lung, gums, and GI tract.[1] In these tissues, modified proteins are produced by biochemical reactions such as citrullination. The mechanism behind environment-triggered RA is thought to be due to the repeated activation of innate immunity. Using the lungs as an example, cigarette smoking induces peptidyl arginine deiminase (PAD) expression in alveolar macrophages, which leads to the conversion of arginine to citrulline in the airway.[41] This process creates a "neoantigen" that activates an immune response and leads to the formation of anti-citrullinated protein antibodies (ACPAs).[41][22]

Patients are genetically predisposed to develop an immune response to the modified proteins, anti-modified protein antibodies (AMPA). Citrullination is produced by the action of PAD on arginine. Isoforms PAD2 and PAD4 are most strongly implicated in RA. Anti-PAD4 antibodies are found in patients with RA, are highly specific for RA, and are associated with ACPA positivity.[42] In other mucosal areas, the inflammatory response in gingivitis is associated with the presence of *Porphyromonas gingivalis*, which causes an inflammatory reaction. There is an influx of leukocytes that produce PAD4. *P. gingivalis* also produces a PAD-like enzyme. These enzymes can citrullinate resident proteins, which then can act as autoantigens.[43] Anti-acetylated protein antibodies have recently been associated with RA (in approximately 40% of RA patients), predominately in seropositive patients.[22] Acetylation is an enzymatic process converting lysine to acetyllysine, thought to be mediated by bacteria, which may provide a link to RA and microbiome dysbiosis. The exact mechanism at this time remains unclear.[22] So there are several types of modified proteins that can be the target of autoantibodies produced by citrullination, carbamylation, and acetylation.[44][45]

Autoantibodies appear before the onset of clinical arthritis.[46][40] Autoimmunity starts at the molecular and cellular level long before the clinical phase of RA, often referred to as pre-symptomatic or pre-clinical RA. During this phase, certain immunologic and biochemical abnormalities have been found, but the patients are asymptomatic.[47][48][46] RF and ACPA can be present in the serum up to 10 years before the onset of clinical symptoms. With time the concentration of ACPA and serum cytokine levels increase.[49][22] Many patients develop autoantibodies but do not develop the overt disease. Some patients will eventually transition from autoimmunity to immune-mediated inflammation primarily focused in the synovium. [1] These autoantibodies are produced by plasma cells in the synovium. The synovium in RA is infiltrated by immune cells, which include innate immune cells (monocytes, dendritic cells, mast cells) and adaptive immune cells (T-helper 1, Th1); T-helper 17, Th17), B cells, and plasma cells). Synovial fibroblast-like synovial cells (FSC) are activated. Neutrophils are not present in the synovium but egress from the blood to the synovial fluid.[50]

Cytokines and chemokines such as tumor necrosis factor (TNF), interleukin-6 (IL-6), and granulocyte-monocyte colony-stimulating factor (GM-CSF) activate endothelial cells and attract immune cells within the synovial compartment.[22] The FSC in the rheumatoid synovium changes to an invasive phenotype. FSC and inflammatory cells produce RANKL leads to osteoclast generation resulting in bone erosions, the hallmark feature of rheumatoid arthritis. [51] It is important to note that synovial biopsies in seropositive patients with arthralgia were essentially unremarkable.[52] It is theorized that a second environmental trigger is needed to cause clinically apparent disease. When this is established, a destructive inflammatory process begins. Fibroblast-like synoviocytes (FLS) migrate from joint to joint, leading to progressive joint damage.[51]

All elements of the immune system are involved, including innate immunity and adaptive immunity, which includes the cellular (T cell) immune response and the humeral (B cell) immune response.[53][45] The innate immune response includes macrophages which produce TNF, IL-6, IL-1, GM-CSF, IL-15, IL-18, IL-32, and chemokines which promote tissue inflammation.[53] Other elements of innate immunity include cells such as endothelial cells, which allow the egress of immune and inflammatory cells such as neutrophils which migrate to the synovial fluid and produce prostaglandins, proteases, and reactive oxygen intermediates, all of which are pro-inflammatory and cause destruction to the cartilage.[53][50]

Resident fibroblasts-like synoviocytes (FLS) are activated by proinflammatory cytokines, PGDF, and chemokines to proliferate with invasive characteristics and produce matrix metalloproteinases (MMP) and ADAMTS. Chondrocytes are activated by TNF, IL-1, and IL-17, producing matrix enzymes resulting in cartilage damage. Resident mast cells produce vasoactive amines, leukotrienes, proteases, and TNF. Osteoclasts are activated by RANKL, which plays a vital role in producing bone erosions.[50] Another element of innate immunity is the complement system which is activated by RF and ACPA-containing immune complexes producing C3a and C5a, which are pro-inflammatory, attracting and activating neutrophils and also activating macrophages and mast cells.[50]

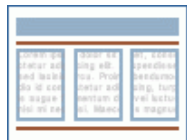
Adaptive immunity includes T cells and B cells which respond to antigenic stimuli. In RA, one of the main antigens triggering autoimmunity is modified proteins such as citrullinated proteins. [1][53][50][54][44][45][55] At mucosal sites, T cells are activated by these modified proteins, which trigger B cells to produce AMPA, including ACPA. In some patients, this immune response develops into an immune-mediated inflammation in the synovium.[53][50] T cells and B cells take up residence in the synovium populating the subsynovial region. The mononuclear cell infiltrate appears histologically like a lymphoid organ. Dendritic cells serve as a bridge between innate and adaptive immunity by presenting antigens to antigen-specific T cells.[53] [50]

T cells interact with B cells, activating antigen-specific B cells to differentiate into plasma cells that produce RF and ACPA. These autoantibodies bind to their antigens forming immune complexes that can further the inflammatory response by activating the complement system.[50][56] Some activated T cells become type 1 helper T cells (Th1) and type 17 helper T cells (Th17). Th17 cells produce IL-17, a proinflammatory cytokine that attracts other T cells and neutrophils.[53][50]

Cytokines are produced in large amounts in the synovium and play an important role in propagating an intense inflammatory response resulting in bone erosions and cartilage loss.[50][53][57] Chronic inflammation also plays a role in associated comorbidities, such as the increased risk of cardiovascular disease. The following table outlines the most important cytokine involved in the pathogenesis of RA.

Cytokines and Chemokines (CXC) Involved in the Pathogenesis of RA

FLS = fibroblast-like synoviocytes; MCP-1 = macrophage chemoattractant protein-1; ILC3 = innate lymphoid cell group 3; MMP = matrix metalloproteinase; NO = nitric oxide; PG = prostaglandins[50]



Table

FLS activation and proliferation, resulting in pannus, activation of chondrocytes, osteoclasts, and endothelial cells.

Histopathology

One of the earliest histopathologic findings in RA is new synovial blood vessel growth; this leads to the transmigration of lymphocytes and polymorphonuclear leukocytes into the synovial fluid. Angiogenesis is required to support the highly catabolic synovium and is accomplished via proinflammatory cytokines such as tumor necrosis factor (TNF).[58] With angiogenesis, cytokines activate endothelial cells to produce adhesion molecules which in turn facilitate cell migration into the synovium. Despite angiogenesis, RA synovial fluid is a hypoxic environment, leading to increased production of cyclooxygenase (COX) 2-derived nociceptive eicosanoids and matrix metalloproteinases (MMPs).[59] This further stimulates an inflammatory response in the synovium. During the early phase of the disease, the influx of inflammatory cells into the synovial membrane leads to a proliferation of monocytes and thickening of the synovial membrane with small villous projections into the joint space.[58]

The histopathology of rheumatoid synovium reviews intense FLS activation and proliferation and subsynovial accumulation of T cells, including Th1, Th17, and Treg cells. B cells also accumulate and differentiate into plasma cells. The appearance of synovial tissue can appear similar to secondary lymphoid tissues such as lymph nodes. Macrophages, dendritic cells, mast cells, and fibroblasts are also present in the subsynovial tissue. Rheumatoid synovial tissue becomes invasive, eroding into the bone and causing marginal erosions.[2]

Rheumatoid nodules initially have a small vessel vasculitis phenomenon followed by a chronic inflammatory granulomatous phase. The development of rheumatoid nodules is frequent in rheumatoid arthritis, involving the dermis or subcutis in peri-articular areas. When multiple, this phenomenon can be related to methotrexate therapy and is called accelerated rheumatoid nodulosis.[60] The size is variable and can range from a few millimeters to centimeters. It appears grossly as a whitish fibrous lesion with yellowish areas corresponding to collagenous necrobiosis. The histological appearance can be indistinguishable from granuloma annulare. There are areas of irregular geographic-like necrobiosis of the dermis and hypodermis surrounded by histiocytes arranged in a well-developed palisade, occasional lymphocytes, and neutrophils. Occasionally, giant cells and mast cells can be present. Fibrin and collagen are present in the center of the necrobiotic areas. The surrounding dermis and

hypodermis have a perivascular infiltrate of plasma cells.[60] Very occasionally, rheumatoid nodules may occur in the lungs.

History and Physical

The most common and predominant symptoms include joint pain and swelling, usually starting insidiously over a period of weeks to months. Initial joint involvement is typically in the small joints of the hands and feet, followed by larger joints. One of the hallmarks is stiffness, particularly morning stiffness.[2][61] Usually, the onset of symptoms is slow and insidious; however, in some cases, an episodic pattern of symptoms can be seen and is defined as palindromic rheumatism.[62] Not all patients with palindromic rheumatism develop RA, and they respond to hydroxychloroquine which supports the idea that palindromic rheumatism is a distinct phenotype of RA.[63]

As discussed above, RA involves small peripheral joints, usually of the hands. The axial joints can be involved, but this is limited to the cervical spine as there are synovial joints in this part of the spine. The lumbar spine is not involved.[64] Although multiple small joint involvements are a frequent feature, some patients may present with monoarticular and extraarticular involvement (lungs).[4] RA is a chronic progressive disease, and without treatment with disease-modifying anti-rheumatic drugs (DMARDs), patients will develop joint damage, deformities, disability, and increased mortality.[4]

On physical examination, the affected joint will be painful if pressure is applied to the joint or on movement with or without joint swelling. Synovial thickening with a "boggy" feel on palpation will be noted. Joint erythema and warmth are usually absent, and wrist involvement may present with typical signs and symptoms of carpal tunnel syndrome. With multiple joint involvements, reduced grip strength will be seen on physical testing. The classical physical findings of ulnar deviation, metacarpophalangeal joint subluxation, ulnar deviation, swan neck deformity, Boutonniere deformity, and the "bowstring" sign are seen in advanced chronic disease. Other findings in the later stages of RA include a reduced range of motion in the shoulders, elbows, and knees. In the feet, hallux valgus is common, as well as other toe deformities.[4]

Rheumatoid nodules are the most common extraarticular manifestations of RA.[65] They are commonly found on pressure points such as the olecranon but may also occur over the joints of the hands and feet, the patella, and the Achilles tendons.

Interstitial lung disease (ILD) affects 5 to 16% of patients with RA and is associated RA specific autoantibodies and increased mortality.[66][67][68] Some patients may present with ILD before developing joint inflammation with positive RF and ACPA. RA-associated ILD is associated with ACPA, anti-carbamylated protein antibodies (anti-CarP), and anti-malondialdehyde-acetaldehyde antibodies (anti-MAA).[69][70][71] The *MUC5B* promoter variant rs35705950 is associated with ILD in patients with RA. This gene encodes mucin 5B, and this variant is a gain of function. RA-ILD compared to unaffected controls, the odds ratio is 3.8 (95% CI 2.8-5.2), and RA-ILD compared to RA without ILD, the odds ratio is 3.1 (95% CI 1.8-5.4).[72]

Ophthalmological manifestations include secondary Sjogren syndrome with dry eyes and also dry mouth.[73] Also, inflammatory eye diseases such as episcleritis and scleritis have become uncommon with new therapies for RA. Similarly, Felty syndrome, which is the combination of long-standing seropositive RA (including RF and ANA), leukopenia, and splenomegaly, is rarely seen today. These patients may have chronic non-healing ulcers and an increased risk of bacterial infection. Vasculitis involving both medium and small-sized blood vessels can also be seen as a presenting symptom of RA. However, the prevalence of vasculitis among patients with RA is low.[74] Patients with rheumatoid vasculitis may exhibit mononeuritis multiplex or asymmetric polyneuropathy.

Evaluation

Laboratory testing often reveals anemia of chronic disease and thrombocytosis. Neutropenia may be present if Felty syndrome is present (rare). RF is present in 80-90% of patients with RA, with a sensitivity of 69% and a specificity of 85%. ACPA is present in 70-80% of patients, with a sensitivity of 67% and a specificity of 95%.[75] Patients with RA with RF, ACPA, or both are designated as having seropositive RA. About 10% of RA patients are seronegative, meaning they do not have RF or ACPA. The presence of rheumatoid factor is not diagnostic of rheumatoid arthritis. It may be present in other connective tissue diseases such as systemic lupus erythematosus, Sjogren syndrome, scleroderma, and granulomatosis with polyangiitis. RF may also be found in patients with chronic infections and healthy individuals, albeit in low titers; if both RF and ACPA are positive, the sensitivity and specificity of the diagnosis increase substantially. RF and ACPA are present for months to years before the onset of clinical arthritis. [46]

Therefore, when a patient presents with inflammatory arthritis, serological studies should be positive. In patients who are asymptomatic or those that have arthralgias, a positive RF and especially CCP predicts the onset of clinical RA.[76][77][78] Healthy relatives of patients with RA who have a positive RF and even more so with a positive ACPA are at increased risk for developing RA.[79][80] Acute phase reactants, such as erythrocyte sedimentation rate (ESR) and levels of C-reactive protein (CRP), are usually elevated in patients with active disease and should be obtained and can be used to assess disease activity when following patients with RA.

Synovial fluid examination usually reveals a leukocyte count between 1500 to 25,000/cubic mm and is predominantly polymorphonuclear cells. Cell counts higher than 25000/cubic mm are rare and can be seen with very active disease; however, they warrant workup to rule out underlying infection. The synovial fluid in RA will also reveal low C3 and C4 levels despite elevated serum levels.[81]

With advanced disease, joint involvement on plain radiographs will reveal periarticular osteopenia, joint space narrowing, and bony erosions (marginal). Erosions of cartilage and bone are considered pathognomonic findings for RA, although not absolutely specific. However, these findings are consistent with advanced disease.[82] Magnetic resonance imaging (MRI) and ultrasonography are useful in early disease before radiographic evidence of bone erosion occurs. [83] A decreased signal from the bone marrow on T1-weighted images and gadolinium-enhanced images indicates bone marrow edema. MRI can also reveal synovial thickening, which has been shown to predict the future presence of bony erosions.[84] The clinical utility of MRI and its incorporation into the diagnostic criteria for RA remains to be determined.

Due to the varied clinical presentation and lack of universal pathognomonic testing for RA, diagnosing the disease can be challenging early in the course of the disease. Traditionally the presence of at least four of the following criteria for at least six weeks would classify the patient as having RA. These criteria were: morning stiffness, arthritis of three or more joints, arthritis of the hands, symmetric arthritis, elevated acute phase reactants, elevated rheumatoid factor, and radiologic evidence of RA. These criteria separated inflammatory from non-inflammatory arthritis but were not very specific for RA. It was also not sensitive to early-stage RA, which was a significant drawback.[85] With the development of serologic markers, the diagnostic criteria were redefined. The 2010 American College of Rheumatology and the European League Against Rheumatism (ACR/EULAR) diagnostic criteria for RA are outlined below. It includes four different domains, which are as follows:

2010 ACR/EULAR Classification Criteria for RA

- Number and size of involved joints[86]
 - 2 to 10 large joints = 1 point (shoulders, elbows, hips, knees, and ankles)

- 1 to 3 small joints = 2 points (metacarpophalangeal joints, proximal interphalangeal joints, second through fifth metatarsophalangeal joints, thumb interphalangeal joints, and wrists)
- 4 to 10 small joints = 3 points
- Greater than 10 joints (including at least 1 small joint) = 5 points
- Serological testing for rheumatoid factor or anti-citrullinated peptide/protein antibody[86]
 - Low positive = 2 points
 - High positive = 3 points
- Elevated acute phase reactant (erythrocyte sedimentation rate [ESR] or C-reactive protein [CRP]) = 1 point[86]
- Symptom duration of at least six weeks = 1 point[86]

A total score greater than or equal to 6 classifies the patient as having RA. Note that these are classification and diagnostic criteria, so their primary purpose is being used in clinical studies of patients with RA, including clinical trials. It is important to note that joint involvement refers to any swollen or tender joint on examination. Imaging studies may also be used to determine the presence of synovitis/joint involvement. The 2010 ACR/EULAR criteria excluded distal interphalangeal joints, first carpometacarpal joints, and first metatarsophalangeal joints from this criteria. Also, these criteria may only be applied to those patients where the joint involvement is not better explained by other inflammatory diseases, such as systemic lupus erythematosus or psoriasis. Specific testing must be obtained to rule out these diseases. The new criteria were noted to better predict the probability of RA, have the same sensitivity as the previous criteria for the diagnosis of RA and have a higher specificity as well as a higher negative predictive value. [85]

Treatment / Management

The goal of treatment in patients with RA is early diagnosis and early initiation of treatment to prevent irreversible damage to the joints. The International Task Force Guidelines published in 2014 make the following recommendations regarding the treatment of RA:[87]

- The primary goal of treatment is to achieve long-term clinical remission and optimize quality of life with the absence of signs and symptoms associated with inflammatory disease activity.
- If clinical remission cannot be achieved, low disease activity is an acceptable alternative.
- Disease activity should be assessed every month in patients with moderate to severe disease activity.
- In patients with low disease activity or clinical remission, disease activity should be assessed every 3 to 6 months.

Multiple clinical assessment tools have been developed to assist clinicians in determining the disease activity of patients with RA. An updated recommendation from the American College of Rheumatology (ACR) in 2019 recommended using the following assessment tools because they met the minimum standard for evaluation per their recommendation.[88][89] In clinical practice, the combination of the DAS28, CDAI, and RAPID3 is easy to use and includes patient input, provider input, and laboratory tests.

- Clinical Disease Activity Index (CDAI)
- Disease Activity Score (DAS)

- Disease Activity Score 28 Joints (DAS28-ESR/CRP)
- Patient-Derived DAS28
- Hospital Universitario La Princesa Index (HUPI)
- Multi-Biomarker Disease Activity Score (MBDA score, VECTRA DA)
- Rheumatoid Arthritis Disease Activity Index (RADAI)
- Rheumatoid Arthritis Disease Activity Index 5 (RADAI-5)
- Routine Assessment of Patient Index Data 3 (RAPID3)
- Routine Assessment of Patient Index Data 5 (RAPID5)
- Simplified Disease Activity Index (SDAI)

Disease-modifying antirheumatic drugs (DMARDs) typically used in treating RA include methotrexate, hydroxychloroquine, sulfasalazine, and leflunomide. Anti-TNF-alpha inhibitors include etanercept, infliximab, adalimumab, golimumab, and certolizumab pegol. Other biologic DMARDs include interleukin (IL) 6 inhibitors such as tocilizumab and sarilumab, T-cell costimulation inhibitors such as abatacept (CTLA4-Ig), and the anti-CD20 B-cell depleting monoclonal antibody such as rituximab. Targeted synthetic DMARDs include Janus kinases (JAK) inhibitors such as tofacitinib, baricitinib, and upadacitinib.[2][4]

DMARD therapy, including biologic agents and targeted synthetic agents (tofacitinib), should be temporarily held in patients with a serious active infection. They can be resumed after the infection has resolved and antimicrobial treatment has been completed.[90] It is essential to remember that all patients starting treatment for RA should be screened for hepatitis B and C and tuberculosis. Methotrexate should be avoided in patients with liver disease.[90] Patients with latent tuberculosis should complete treatment for at least one month before initiating biologic and targeted synthetic agents. If patients cannot take or complete treatment for latent tuberculosis, conventional DMARD therapy should be used.[90] In patients with underlying skin cancer and lymphoproliferative disorders, biologic agents should be avoided except for rituximab in patients with lymphoproliferative disorders, as there is evidence of benefit from B-cell suppression in these cases.[90] The ACR also recommends that before starting therapy for RA, patients should receive vaccination for pneumococcus, hepatitis, influenza, human papillomavirus (HPV), herpes zoster virus (HZV), and COVID-19.

Non-steroidal Anti-inflammatory Drugs (NSAIDs)

NSAIDs do not have any disease-modifying effects but are commonly used to relieve symptoms related to joint inflammation and pain. There are about 20 such drugs (depending on what country you are in) that are all effective at full doses. There is some variation in side effects and toxicities. There is a class Black Box warning for cardiovascular disease, although there is variation in the cardiovascular effects among NSAIDs.[91][92] NSAIDs also have the potential for gastrointestinal, renal, and hematologic toxicity. The choice of a specific NSAID depends on the patient, their comorbidities, and the provider's familiarity with the specific drug. Non-acetylated salicylates (salsalate) have an excellent safety profile with little GI, renal, or hematologic effects but are not commonly used. There are recommendations for using NSAIDs with regard to a patient's cardiovascular and gastrointestinal comorbidities.[93]

- Efficacy: indomethacin, piroxicam
- Convenience (1-2 doses per day): meloxicam, naproxen, celecoxib, nabumetone
- Overall safety: nabumetone[94]
- GI safety: celecoxib, nabumetone

- Renal safety: possible nabumetone
- Hematologic safety (no antiplatelet effect): celecoxib, meloxicam, nabumetone

Corticosteroids

Corticosteroids are commonly used in patients with RA. There are several situations in which corticosteroids should be considered. In a new patient with very active RA, corticosteroids can be used as bridge therapy while DMARD therapy is instituted. Some studies show that using corticosteroids early in RA patients improves outcomes and has disease-modifying effects, including radiographic progression.[95][96] Another study did not show long-term benefits from corticosteroid bridge therapy.[97] Short courses of corticosteroids can be used for mild flares of RA. Intra-articular corticosteroids can be used for single-joint flares. About 50% of patients with RA require low-dose corticosteroids (prednisone 2.5 to 7.5 mg daily to maintain control of their disease. Long-term corticosteroids are associated with numerous toxicities, including weight gain, osteoporosis, and increased risk of infections.

Nonbiologic DMARDs

This category includes methotrexate, hydroxychloroquine (HCQ), azathioprine (AZA), sulfasalazine, leflunomide, and cyclosporine. Methotrexate is the initial drug of choice for patients with RA. The recommended treatment plan recommends an initial dose of 10-15 mg/week of methotrexate with an escalation of 5 mg/month and a target dose of 20-25 mg/week.[98] Subcutaneous administration can be tried in patients who do not respond adequately or are intolerant to oral administration.[99] In patients on methotrexate, 15 mg/week or higher, who still have some disease activity, splitting the dose (10 mg in the morning and 10 mg in the evening every week) increased the bioavailability of methotrexate.[100] Triple therapy, consisting of the combination of methotrexate, hydroxychloroquine, and sulfasalazine, has been shown to be an effective regimen for RA.[101][102] However, triple therapy is not well tolerated, and a large percentage of patients do stay on this therapy.[103][104][105]

TNF inhibitors

The TNF inhibitors include etanercept, infliximab, adalimumab, certolizumab, and golimumab. The ACR does not recommend using TNF inhibitors until a nonbiologic DMARD has been tried.[90] However, studies have shown that adding TNF inhibitors in patients who have failed methotrexate therapy is better than adding another nonbiologic DMARD.[106] The most concerning adverse effect of these agents is opportunistic infections and reactivation of latent tuberculosis. There is some concern about the generation of antibodies against these agents, which may decrease their efficacy over time; however, using methotrexate in combination with these agents has been shown to decrease this complication.[107] Acute infection, advanced heart failure, demyelinating disease, and recent malignancies are contraindications for using these agents.[90]

Rituximab

Rituximab is a biologic DMARD that can be added for treating RA if patients have uncontrolled RA and who did not respond to TNF inhibitors.[108] Rituximab is given as an intravenous infusion; it depletes CD20+ B-cells and decreases the immune response to vaccines in patients receiving rituximab. Responses to rituximab are better if the patients are seropositive and if the patients are also on methotrexate. As stated above, it is a preferred agent in patients with underlying lymphoproliferative disorders.

Abatacept

Abatacept inhibits T-cell activation by binding to CD80 and CD86. It is administered as a monthly intravenous infusion or as a weekly subcutaneous injection. Patients with uncontrolled

RA, who have shown an inadequate response to methotrexate and TNF inhibitor therapy, benefit from abatacept therapy with proven efficacy from 6 months to 5 years of therapy.[109]

Interleukin 6 Inhibitors

Tocilizumab, an IL-6 receptor inhibitor, is indicated for moderate-to-severe active RA in adults who have had an inadequate response to TNF inhibitor therapy. These patients develop clinically meaningful improvement with the use of tocilizumab.[110] Sarilumab is another agent in this category that has been shown to improve clinical outcomes in patients with uncontrolled RA despite TNF inhibitor therapy.[111]

Janus kinase (JAK) Inhibitors

JAK is a group of tyrosine kinases that participate in intracellular signal transduction for hematopoiesis and immune cell function. JAK inhibitors (such as tofacitinib) are oral agents that reduce the production of cytokines and are approved as second-line agents for the treatment of RA.[112]

ACR and EULAR Treatment Guidelines

- According to the ACR treatment guidelines for early RA, patients who have not taken disease-modifying antirheumatic drug (DMARD) therapy should start DMARD therapy regardless of the disease activity level.
- In patients with low disease activity and early disease, monotherapy with methotrexate is the preferred treatment.
- Leflunomide or sulfasalazine are the first-line treatment in patients with a contraindication to methotrexate or intolerance to it.
- If monotherapy with DMARD does not control disease activity (regardless of concomitant glucocorticoid use), therapy should be altered. Methotrexate can be continued or discontinued at this point. Additional therapy options after failed monotherapy with DMARD are recommended as either dual traditional/nonbiologic DMARD therapy, tumor necrosis factor (TNF) inhibitors, or non-tumor necrosis factor biologic agents.
- In patients with established RA, who are DMARD naive, methotrexate is the preferred agent for initial monotherapy, regardless of the disease activity level.
- If monotherapy with DMARD does not control disease activity in established RA (regardless of concomitant glucocorticoid use), dual DMARD therapy, a TNF inhibitor, a non-TNF biologic agent, or tofacitinib therapy can be added.
- If disease activity remains high on TNF inhibitor monotherapy, DMARD therapy should be added in addition to the TNF inhibitor.
- If disease activity remains high despite a TNF inhibitor switch to a non-TNF biologic agent with or without methotrexate
- If disease activity remains high despite a trial of anti-TNF and non-TNF agents, use another non-TNF biologic agent before considering tofacitinib.
- If still uncontrolled despite the above trials, use tofacitinib
- If disease activity remains high despite the above combination therapies, short-term low-dose glucocorticoid therapy should be added.
- TNF inhibitors should be avoided in patients with congestive heart failure.
- Patients with hepatitis C who have not been treated or are currently not on treatment for it should receive nonbiologic DMARD therapy rather than TNF inhibitors.[90]

Safety Considerations with Biologic and Targeted Synthetic DMARDs

- Serious infections: All biologic and targeted synthetic DMARDs are associated with 2 to 3 times the number of infections. Hold all agents during an active infection, hold for surgical procedures, and avoid live vaccines. Opportunistic infections, including TB and fungal infections, all patients should be assessed for latent TB.
- Herpes zoster: JAK inhibitors
- Neutropenia: IL-6 inhibitors, JAK inhibitors, rituximab
- Elevated LFTs, elevated cholesterol: IL-6 inhibitors, JAK inhibitors
- Thrombosis: JAK inhibitors
- CHF: TNH inhibitors
- Cardiovascular events: JAK inhibitors (Black Box warning)
- Demyelinating diseases: TNF inhibitors

Differential Diagnosis

- Osteoarthritis
- Psoriatic arthritis
- Systemic lupus erythematosus
- Sjogren syndrome
- Polymyalgia rheumatic
- Chronic gouty arthritis
- Calcium pyrophosphate deposition disease

Staging

Stages of RA as Defined by the ACR

- Stage 1: No destructive changes on X-rays
- Stage 2: Presence of X-ray evidence of periarticular osteoporosis, subchondral bone destruction but no joint deformity
- Stage 3: X-ray evidence of cartilage and bone destruction in addition to joint deformity and periarticular osteoporosis
- Stage 4: Presence of bony or fibrous ankylosis along with stage 3 features^[90]

Prognosis

Rheumatoid arthritis has no cure and is a progressive disease associated with morbidity and increased mortality. All individuals will experience multiple exacerbations and, without treatment, tend to have poor outcomes with disability and increased mortality.^{[113][5][113]} Early treatment (within six months of symptom onset) has shown improved functional capacity and decreased disease activity as measured by swollen joint count and tender joint count. However, the mortality rate was similar in patients receiving early treatment and late treatment (after six months of symptoms), and both were significantly improved from no treatment.^[113] Approximately 40% of patients with RA will have a functional disability affecting their ability to work and complete activities of daily living within ten years of the diagnosis.^[114]

Patients with RA also develop other chronic medical conditions/complications which drastically affect their outcomes. The most notable of this correlation is the relationship between RA and atherosclerotic cardiovascular disease leading to accelerated coronary artery disease.[115] [116] RA increases the risk of cardiovascular disease, lung diseases, and malignancies, which in turn increases the risk of premature mortality in these patients.[117]

Complications

Complications of RA span multiple organ systems and are known to worsen clinical outcomes in patients with RA. It is imperative to monitor patients for the development of these complications and quickly alter treatment plans if applicable. Frequent recurrent serious opportunistic infections occur in patients with RA, which warrants withholding DMARD therapy until they are treated. The increased frequency of infections in patients with RA is thought to be due to underlying immune dysfunction from the disease itself and the use of DMARD therapy.[118]

Complications and Comorbidities Associated with Uncontrolled Systemic Inflammation

- Chronic joint inflammation results in radiographic damage, progressive deformity, and functional disability.
- Anemia of chronic disease and Felty syndrome are well-documented complications of seropositive RA.[73]
- The secondary form of Sjögren syndrome is associated with RA and can have a prevalence of as high as 10% in patients with RA and pulmonary disease.[73] Other complications are uncommon to rare, including inflammatory eye disease and Felty's syndrome. Rheumatoid vasculitis is a rare phenomenon but can have severe complications if present. Clinical presentation ranges from focal digital involvement to severe systemic involvement resembling polyarteritis nodosa.[74]
- Pleuritis, bronchiolitis, and interstitial lung disease are also associated with RA. Although rare, RA treatment with methotrexate and anti-TNF agents can also lead to pulmonary injury. There is also an increased risk of pulmonary embolism in RA. Interstitial lung disease (ILD) affects 5 to 16% of RA patients and is associated with RA-specific autoantibodies and increased mortality.[66][67][68] Some patients may present with ILD before developing joint inflammation with positive RF and ACPA. RA-associated ILD is associated with ACPA, anti-carbamylated protein antibodies (anti-CarP), and anti-malondialdehyde-acetaldehyde antibodies (anti-MAA).[69][70][71] The *MUC5B* promoter variant rs35705950 is associated with ILD in patients with RA. This gene encodes mucin 5B, and this variant is a gain of function. RA-ILD compared to unaffected controls, the odds ratio is 3.8 (95% CI 2.8-5.2), and RA-ILD, compared to RA without ILD, the odds ratio is 3.1 (95% CI 1.8-5.4).[72]
- Coronary artery disease has a strong association with RA.[116] RA is an independent risk factor for the development of coronary artery disease (CAD) and accelerates the development of CAD in these patients.[115] Accelerated atherosclerosis is the primary cause of morbidity and mortality in patients with RA leading to CAD and peripheral vascular disease.[119] There is increased insulin resistance and diabetes mellitus associated with RA, and is thought to be due to chronic inflammation. When treated with specific DMARDs such as hydroxychloroquine, methotrexate, and TNF antagonists, there was a marked improvement in glucose control in these patients.[120] The use of conventional and biologic DMARDs is associated with a reduced risk of cardiovascular events.[121] TNF inhibitors and abatacept appear to have the greatest effect on cardiovascular disease, and the addition of methotrexate to biologic agents is associated with an even lower risk of cardiovascular events.[122][123]

- Patients with RA also have a greater risk of developing lymphoma, with a higher incidence of non-Hodgkin lymphoma in these patients. The clinical course of RA in these patients is accelerated, and diffuse B-cell lymphoma is often the most common subtype.[124] Non-Hodgkin's lymphoma is more common in patients with RA, but this occurs mainly in patients with persistently active disease.[125] The use of biologic DMARDs does not appear to increase the risk of lymphoma.[126]

Complications and Comorbidities Associated with RA Disease and Treatment

- Premature death
- Serious infections
- Osteopenia and osteoporosis are complications of the disease itself and can also be associated with drug therapies (glucocorticoids). Compared to the general population, patients with RA have a 60% to 100% increased risk of fracture.[127] Patient factors that increase the risk of this complication in patients with RA are the same as those of osteoporosis, including postmenopausal state, low body mass index, and advanced age.
- There is an increased risk of venous thromboembolic disease in patients with RA, as stated above, even after adjusting for other risk factors for thromboembolic disease. [128] Multiple studies have reported a higher risk of thromboembolic disease in patients receiving TNF inhibitor therapy and JAK inhibitors.[129][130] however, the consensus regarding this association is not well established. Some studies suggest that higher thromboembolic disease with the use of these agents is secondary to higher disease activity rather than an adverse effect of the agents.[130]

Others

- Depression is a significant complication of RA. It is present in patients with long-term active disease and debilitating physical dysfunction. A 2013 meta-analysis reported a 17% to 39% prevalence of depression in patients with RA.[131]

Deterrence and Patient Education

RA has a strong genetic component and cannot be entirely avoided; however, modifiable risk factors such as cigarette smoking and consuming a balanced diet should be addressed. Patients at risk for RA should be counseled to eat a well-balanced diet, exercise regularly, and maintain healthy body weight.

Enhancing Healthcare Team Outcomes

Rheumatoid arthritis is a chronic disorder with no cure. It affects many other organs and is best managed with an interprofessional team to provide comprehensive care. The key is patient education by nurses, pharmacists, rheumatologists, and primary care providers. The nurse should inform the patient about the signs and symptoms of different organ systems and when to seek medical care. The patient should enroll in an exercise program to recover joint function. An occupational therapy consult can help the patient manage daily living activities. The pharmacist should educate the patient on the types of drugs used to treat rheumatoid arthritis and their potential side effects while checking for interactions and reporting their findings to the team for therapy modification, if necessary.

At each clinic visit, clinicians and nurses should educate the patient to stop smoking, maintain a healthy body weight, get the recommended vaccinations and eat a healthy diet. The social worker should be involved in ensuring that the patient's home is liveable and the patient has ample support systems. The patient should be encouraged to join support groups. Due to progressive disability, chronic pain, and poor quality of life, many patients become depressed and

need a mental health nurse and counselor to cope with their disease. A comprehensive interprofessional team can provide a multi-faceted approach to treating RA and improving patient outcomes. [Level 5]

RA has frequent relapses and remissions. While some patients have mild disease, others may have a disease that severely affects their quality of life. Worse outcomes are usually seen in patients with a high titer of autoantibodies, HLA-DRB1 genotypes, age younger than 30 years at onset, multiple joint involvements, female gender, and extra-articular involvement on presentation. In addition, the drugs used to treat rheumatoid arthritis also have potent side effects, which often are not well tolerated. As the disease progresses, many patients will develop adverse cardiac events leading to death. The overall mortality in patients with rheumatoid arthritis is three times higher than in the general population.[132][133] A comprehensive interprofessional team approach to treating and monitoring patients with RA can help improve clinical outcomes for these patients. [Level 5]

Review Questions

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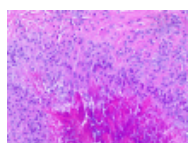
Figure

Summary keynotes on Rheumatoid Arthritis Contributed by StatPearls



Figure

Ultrasound image of the thumb in a patient with a history of Rheumatoid Arthritis shows an abnormal amount of fluid within the flexor pollicis longus tendon sheath, consistent with an inflammatory tenosynovitis. Contributed by Mike Tall, MD



Figure

Granuloma annulare, H/E 20x. Palisading histiocytes and lymphocytes surround an area of necrobiotic collagen. Contributed by Fabiola Farci, MD

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Disclosure: Krati Chauhan declares no relevant financial relationships with ineligible companies.

Disclosure: Jagmohan Jandu declares no relevant financial relationships with ineligible companies.

Disclosure: Lawrence Brent declares no relevant financial relationships with ineligible companies.

Disclosure: Mohammed Al-Dhahir declares no relevant financial relationships with ineligible companies.

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